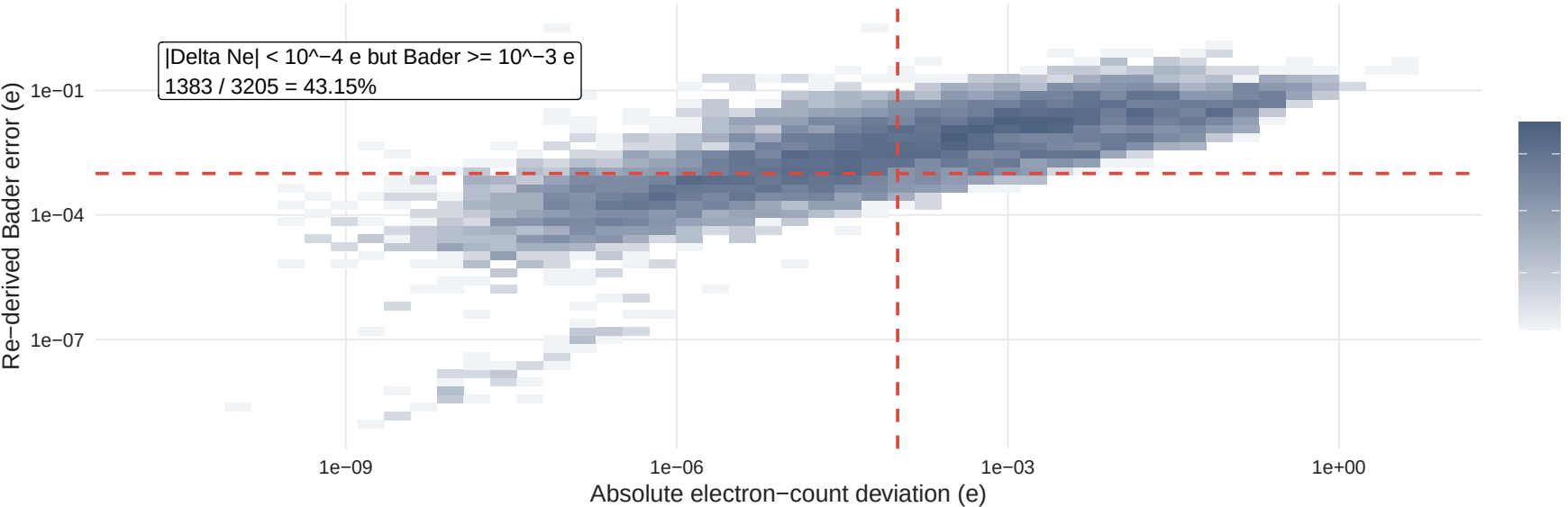


Supplementary Figure S3 | Extended controls separate global, smooth nonlocal and topology-sensitive QoIs

The same frozen reconstruction corpus is interrogated with total electron count, periodic Hartree potential and re-derived Bader charge.

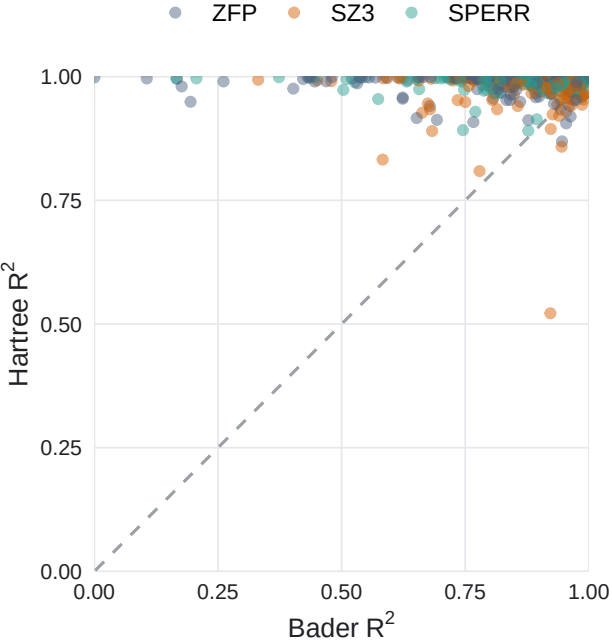
A | Global electron-number conservation is not a local chemical certificate

All finite development reconstructions; dashed lines mark the negative-control thresholds



B | Hartree response is smoother on the

Paired material-codec fits; n = 678 pairs



C | Strict monotonicity shows a bulk/slab nuance

Hartree remains smoother by R^2 , but slab ladders contain small local Hartree wiggles



D | Similar Hartree error can coexist with order-of-magnitude Bader dispersion

0.5-decade Hartree-error bins with at least 10 rows

